EXERCISE 05: The quenched solution to randomised perceptron learning

Consider a student perceptron with N binary input elements and one binary output receiving the classification results for $P = \alpha N$ random examples from an identical teacher perceptron. According to the quenched computation, the typical student achieving zero training error also has an overlap $R(\alpha)$ with the teacher such that

$$\frac{R}{\sqrt{1-R}} = \frac{\alpha}{\pi} I(R,\alpha) \ , \tag{1}$$

with

$$I(R,\alpha) = \int_{-\infty}^{\infty} \frac{dv}{\sqrt{2\pi}} \frac{\exp(-(1+R)v^2/2)}{H(-\sqrt{R}v)} \ ,$$

and

$$H(u) = \int_{u}^{\infty} \frac{dx}{\sqrt{2\pi}} \exp(-x^2/2) \ .$$

Recall that the teacher student typical overlap R determines the typical student's generalisation error $\epsilon(R) = \frac{1}{\pi} \arccos(R)$.
Superimpose to the numerical results obtained in Exercise 03 the result of equation (1) obtained through the following steps:

- suitably choose a parameter t (and state explicitly your choice) to represent the two dimensional curve $\epsilon(\alpha)$ by means of a parametric plot;
- consider a strategy to solve explicitly equation (1) in correspondence of a given α by setting up an iterative procedure:
  - rewrite the equation as $R = f(R,\alpha)$
  - start with R_0
  - evaluate $R_{i+1} = (1-a)R_i + af(R_i,\alpha)$
  - plot R_i *vs* i to check if it converges
- check that possible choices of $f(R,\alpha)$ are either

  $$f_1(R,\alpha) = \frac{\alpha}{\pi}\sqrt{1-R}\ I(R,\alpha)$$

  or

  $$f_2(R,\alpha) = 1 - \frac{\pi^2 R^2}{\alpha^2 I(R,\alpha)^2}$$

  and for both cases show the plot R_i *vs* i for
  - $\alpha = 5.0$, $R_0 = 0.5$, $a = 0.5$
  - $\alpha = 0.1$, $R_0 = 0.5$, $a = 0.5$
- is any of the two choices for $f(R,\alpha)$ suitable on the entire range $\alpha \in [0:\infty]$?
- starting from your understanding of the convergence of the iterative equations, set up an automatic procedure to determine the result $\epsilon(\alpha)$ with high precision in correspondence of many $\alpha \in [0:10]$
- plot the obtained curve $\epsilon(\alpha)$ on top of the parametric plot and of the numerical result from Exercise 03.